MAXWELL EQUATIONS & BOUNDARY CONDITIONS

$\nabla \times \vec{H} = \vec{j}_c + \frac{\partial \vec{D}}{\partial t}$	$\oint \vec{H} \cdot d\vec{l} = \int_S \left(\vec{j}_c + \frac{\partial \vec{D}}{\partial t} \right) \cdot \vec{ds}$ Ampere Law
$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\oint \vec{E} \cdot \vec{dl} = \int_S \left(-\frac{\partial \vec{B}}{\partial t} \right) \cdot \vec{ds}$ Foraday Law
$\nabla \cdot \vec{D} = \rho$	$\oint_S \vec{D} \cdot \vec{ds} = \int_V \rho \, dv$ Gauss Law
$\nabla \cdot \vec{B} = 0$	$\oint_S \vec{B} \cdot \vec{ds} = 0$ absence of monopoles

At the interface of two dielectrics :

Magnetic fields

$$B_{n1} = B_{n2}$$

$$H_{t1} = H_{t2}$$

$$(\vec{H}_1 - \vec{H}_2) \times \vec{a}_{n12} = \vec{j}_s$$

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\mu_{r2}}{\mu_{r1}}$$

Electric fields

$$D_{n1} = D_{n2}$$

$$(\vec{D}_1 - \vec{D}_2) \cdot \vec{a}_{n12} = \rho_s$$

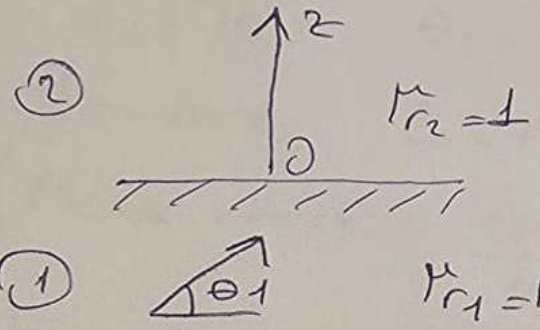
$$E_{t1} = E_{t2}$$

$$\frac{\tan\theta_1}{\tan\theta_2} = \frac{\varepsilon_{r2}}{\varepsilon_{r1}}$$

Ex

In the figure, in region 1, $\vec{B}_1 = 1.2\vec{a}_x + 0.8\vec{a}_y + 0.4\vec{a}_z$ (T)

Find H_2 and determine the angles between the field vectors and the surface tangent.



Solution:

$$\vec{B}_1 = 1.2\vec{a}_x + 0.8\vec{a}_y + 0.4\vec{a}_z \quad (T)$$

$$\vec{H}_1 = \frac{\vec{B}_1}{\mu_1} \Rightarrow \vec{H}_1 = \frac{1}{\mu_0}\left(8.0\vec{a}_x + 5.33\vec{a}_y + 2.67\vec{a}_z\right)10^{-2} \ A/m$$

$$H_{1t} = H_{2t} \Rightarrow \begin{matrix} H_{1x} = H_{2x} \\ H_{1y} = H_{2y} \end{matrix}$$

$$B_{n1} = B_{n2} \Rightarrow B_{2z} = B_{1z} \Rightarrow B_{1z} = 0.4 = B_{2z}$$

$$B_{2x} = \mu_0 \mu_{r_2} H_{2x} \qquad B_{2y} = \mu_0 \mu_{r_2} H_{2y}$$

$$B_{2x} = 8.0 \times 10^{-2} \ T \qquad B_{2y} = 5.33 \times 10^{-2} \ T$$

$$H_{2z} = \frac{B_{2z}}{\mu_0 \mu_{r_2}} = \frac{0.4}{\mu_0} \ (A/m)$$

Since θ_1 is the angle between α_1, $\vec{B}_1$, and the normal $\vec{a}_2$, it is equal to $90^\circ - \alpha_1$

$$\cos\alpha_1 = \frac{\vec{B}_1 \cdot \vec{a}_2}{|B_1|} = 0.27$$

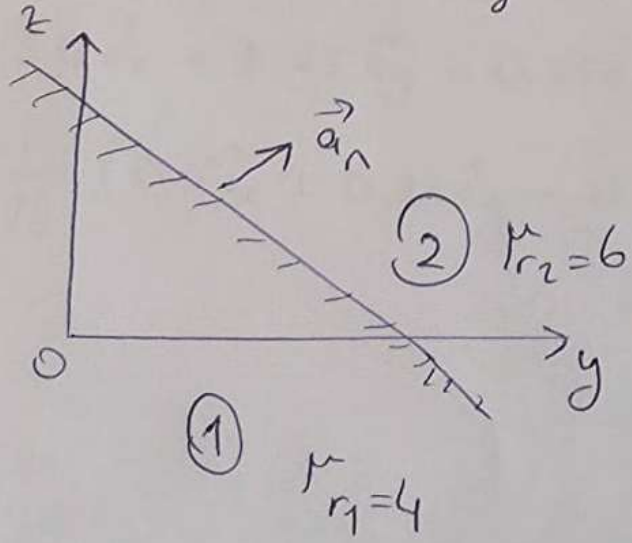
$$\alpha_1 = 74.5^\circ \quad \theta_1 = 15.5^\circ$$

Similarly $\theta_2 = 76.5^\circ$

(Please check: $\frac{\tan\theta_1}{\tan\theta_2} \stackrel{?}{=} \frac{\mu_{r_2}}{\mu_{r_1}}$)

Ex

Region 1 has $\mu_{r_1} = 4$ and is located on the side of the plane $y+z=1$ that contains the origin. In Region 2, $\mu_{r_2} = 6$. If $\vec{B}_1 = 2.0\vec{a}_x + 1.0\vec{a}_y$ (T) Find $\vec{B}_2$ & $\vec{H}_2$.



Solution:

Unit normal $\vec{a}_n = (\vec{a}_y + \vec{a}_z)/\sqrt{2}$

$$B_{n1} = \frac{(2.0\vec{a}_x + 1.0\vec{a}_y)\cdot \vec{a}_y + \vec{a}_z}{\sqrt{2}} = \frac{1}{\sqrt{2}} = B_1 \cdot \vec{a}_n$$

$$\vec{B}_{n1} = \left(\frac{1}{\sqrt{2}}\right)\vec{a}_n = 0.5\vec{a}_y + 0.5\vec{a}_z = \vec{B}_{n2}$$

$$\vec{B}_{t1} = \vec{B}_1 - \vec{B}_{n1} = 2.0\vec{a}_x + 0.5\vec{a}_y - 0.5\vec{a}_z$$

$$\vec{H}_{t1} = \frac{1}{\mu_0}\left(0.5\vec{a}_x + 0.125\vec{a}_y - 0.125\vec{a}_z\right) = \vec{H}_{t2}$$

$$\vec{B}_{t2} = \mu_0\mu_{r_2}\vec{H}_{t2} = 3.0\vec{a}_x + 0.75\vec{a}_y - 0.75\vec{a}_z$$

Now write down $\vec{B}_2 = \vec{B}_{2n} + \vec{B}_{2t}$

$$\vec{B}_2 = 3.0\vec{a}_x + 1.25\vec{a}_y - 0.25\vec{a}_z \ (T)$$

$$\vec{H}_2 = \frac{1}{\mu_0}\left(0.5\vec{a}_x + 0.21\vec{a}_y - 0.04\vec{a}_z\right)(A/m)$$

Ex

A current sheet $\vec{j} = 6.5\vec{a}_z$ A/m at $x=0$ seperates Regron 1 (where $x<0$) with $\vec{H}_1 = 10\vec{a}_y$ A/m from Regron 2 (where $x>0$). Find $\vec{H}_2$ at $x=+0$.

Solution:

Nothing is stated about the permeabilities of the two regrons; however, since $\vec{H}_1$ is entirely tangential, a change in permeability will have no effect.

Because $\vec{B}_{n1}=0$, it follows that $\vec{B}_{n2}=0$, and therefore $\vec{H}_{n2}=0$

$$(\vec{H}_1 - \vec{H}_2) \times \vec{a}_{n12} = \vec{j}$$

$$(10\vec{a}_y - H_{y2}\vec{a}_y) \times \vec{a}_x = 6.5\vec{a}_z$$

$$(10 - H_{y2})(-\vec{a}_z) = 6.5\vec{a}_z$$

$$H_{y2} = 16.5 \text{ A/m}$$

Hence,

$$\vec{H}_2 = 16.5\vec{a}_y \text{ A/m}$$

Ex:

In Region 1 where $z<0$, $\mu_{r_1} = 1.5$

In Region 2 where $z>0$, $\mu_{r_2} = 5$. Near the point $(0,0,0)$, the fields are given as:

$$\vec{B}_1 = 2.40\,\vec{a}_x + 10.0\,\vec{a}_z \ (T)$$

$$\vec{B}_2 = 25.75\,\vec{a}_x - 17.7\,\vec{a}_y + 10.0\,\vec{a}_z \ (T)$$

If there is a surface current at the interface, what is its density at the origin?

Solution:

Near the origin

$$\vec{H}_1 = \frac{1}{\mu_0 \mu_{r_1}} \vec{B}_1 = \frac{1}{\mu_0}\left(1.60\,\vec{a}_x + 6.67\,\vec{a}_z\right) \ A/m$$

$$\vec{H}_2 = \frac{1}{\mu_0}\left(5.15\,\vec{a}_x - 3.54\,\vec{a}_y + 2.0\,\vec{a}_z\right) \ A/m$$

Hence

$$\vec{J} = (\vec{H}_1 - \vec{H}_2) \times \vec{a}_{n12} = \frac{1}{\mu_0}\left(-3.55\,\vec{a}_x + 3.54\,\vec{a}_y + 4.67\,\vec{a}_z\right) \times$$

$$= \frac{5.0}{\mu_0}\left(\frac{\vec{a}_x + \vec{a}_y}{\sqrt{2}}\right) \ A/m$$

Ex

In free space $\vec{E} = E_m \sin(\omega t - \beta z)\, \vec{a}_y$ is given. Find $\vec{D}$, $\vec{B}$ and $\vec{H}$. Plot $\vec{E}$ and $\vec{H}$ at $t=0$.

Solution:

$$\vec{D} = \varepsilon_0 \vec{E} = \varepsilon_0 E_m \sin(\omega t - \beta z)\, \vec{a}_y$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\begin{vmatrix} \vec{a}_x & \vec{a}_y & \vec{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_m \sin(\omega t - \beta z) & 0 \end{vmatrix} = -\frac{\partial \vec{B}}{\partial t}$$

$$-\frac{\partial \vec{B}}{\partial t} = \beta E_m \cos(\omega t - \beta z)\, \vec{a}_x$$

By integrating

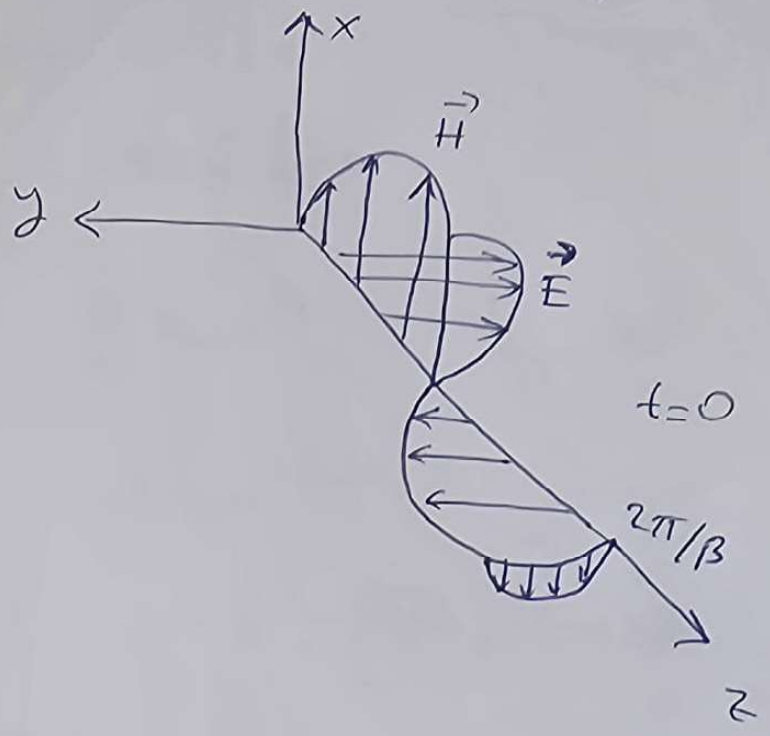
$$\vec{B} = -\frac{\beta E_m}{\omega} \sin(\omega t - \beta z)\, \vec{a}_x$$

$$\vec{H} = -\frac{\beta E_m}{\omega \mu_0} \sin(\omega t - \beta z)\, \vec{a}_x$$

Since $\vec{E}$ and $\vec{H}$ are mutually perpendicular, note that they are orthogonal.

At $t=0$:

$$\sin(\omega t - \beta z) = -\sin(\beta z)$$



$\beta \to k$

Ex

If in free space $\vec{H} = H_m e^{j(\omega t + \beta z)} \vec{a}_x$ is given Find $\vec{E}$.

Solution:

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$\frac{\partial}{\partial z} H_m e^{j(\omega t + \beta z)} \vec{a}_y = \frac{\partial \vec{D}}{\partial t}$$

$$j\beta H_m e^{j(\omega t + \beta z)} \vec{a}_y = \frac{\partial \vec{D}}{\partial t}$$

$$\vec{D} = \frac{\beta H_m}{\omega} e^{j(\omega t + \beta z)} \vec{a}_y$$

$$\vec{E} = \frac{\vec{D}}{\varepsilon_0}$$

=

PLANE EM WAVES

$$\nabla^2 E + k^2 E = 0$$

k: wavenumber

$$k = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{u_p} \quad \text{rad/m}$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right) E_x = 0$$

Uniform plane wave characterized by a uniform E_x over plane surfaces perpendicular to z:

$$\frac{d^2 E_x}{dz^2} + k^2 E_x = 0$$

$$E_x(z) = E_x^+(z) + E_x^-(z) = E_0^+ e^{-jkz} + E_0^- e^{jkz}$$

Examine the first phasor term on the right side

$$\vec{E}(z) = \vec{a}_x E_x^+(z) = \vec{a}_x E_0^+ e^{-jkz}$$

$$\vec{E}(z,t) = \vec{a}_x E_x^+(z,t) = \vec{a}_x \mathrm{Re}\left[E_x^+(z) e^{j\omega t}\right]$$

$$= \vec{a}_x E_0^+ \cos(\omega t - kz)$$

$$u_p = \frac{dz}{dt} = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\varepsilon}}$$

$$k = \frac{2\pi}{\lambda}$$

Second phasor term on the right side $\bar{E}_0 e^{jkz}$ represents a cosinusoidal wave traveling in the $-z$ direction with the same velocity up.

Intrinsic Impedance $$\eta = \sqrt{\frac{\mu}{\varepsilon}} = \frac{E}{H}$$

$$H(z,t) = \vec{a}_y H_y^+(z,t) = \vec{a}_y \, Re\left[H_y^+(z) e^{j\omega t} \right]$$

$$= \vec{a}_y \frac{E_0^+}{\eta} \cos(\omega t - kz)$$

TRANSVERSE EM WAVES

$\vec{E}$ and $\vec{H}$ are perpendicular to each other, and both are transverse to the direction of propagation. It is called TEM waves.

Let's assume:

$$E(x,z) = \vec{a}_y E_0 e^{-jk_x x - jk_z z}$$

y directed $\vec{E}$ of a uniform plane wave propagating in both +x and +z directions.

$$\vec{k} = \vec{a}_x k_x + \vec{a}_z k_z = \vec{a}_k k$$

$\vec{R}$: a radius vector R from the origin to an arbitrary point

$$\vec{R} = \vec{a}_x x + \vec{a}_y y + \vec{a}_z z$$

$$\vec{E} = \vec{a}_y E_0 e^{-j\vec{k}\cdot\vec{R}}$$

$$\vec{H} = -\frac{1}{j\omega\mu} \nabla \times \vec{E} = \frac{k}{\omega\mu} \vec{a}_k \times \vec{E} = \frac{1}{\eta} \vec{a}_k \times \vec{E}$$

POLARIZATION OF PLANE WAVES

The polarization of a uniform plane wave describes the time-varying behavior of the electric field intensity vector at a given point in space.

If $\vec{E} = \vec{a}_x E_x$ (a plane wave is fixed in the x direction)

↓

Linearly polarized in the x direction.

$E(z) = \vec{a}_x E_1(z) + \vec{a}_y E_2(z)$ → superposition of two linearly polarized waves

The instantaneous expression for E is

$$E(z,t) = Re\left\{\left[\vec{a}_x E_1(z) + \vec{a}_y E_2(z)\right] e^{j\omega t}\right\}$$

$$= \vec{a}_x E_{10} \cos(\omega t - kz) + \vec{a}_y E_{20} \cos\left(\omega t - kz - \pi/2\right)$$

The direction change of $\vec{E}$ at a given point as t changes, it is convenient to set $z=0$:

$$E(0,t) = \vec{a}_x E_1(0,t) + \vec{a}_y E_2(0,t)$$

$$= \vec{a}_x E_{10} \cos\omega t + \vec{a}_y E_{20} \sin\omega t$$

ωt completes the cycle from $0 \rightarrow 2\pi$

$E(0,t)$ will traverse an elliptical locus in the counterclockwise direction.

$$\cos\omega t = \frac{E_1(0,t)}{E_{10}} \qquad \sin\omega t = \frac{E_2(0,t)}{E_{20}}$$

$$\sqrt{1-\cos^2\omega t} = \sqrt{1-\left[\frac{E_1(0,t)}{E_{10}}\right]^2}$$

$$\left[\frac{E_2(0,t)}{E_{20}}\right]^2 + \left[\frac{E_1(0,t)}{E_{10}}\right]^2 = 1$$

Sum of the two linearly polarized waves in both space and time quadrature

If $E_{20} \neq E_{10}$ it is elliptically polarized

$E_{20} = E_{10}$ circularly polarized

Ex
7.1

a) Obtain the wave equations governing the $\vec{E}$ and $\vec{H}$ fields in a source-free conducting medium with constitutive parameters ε, μ, and σ.

b) Obtain the corresponding Helmholtz's equations for time-harmonic fields.

Ex
7.3

Obtain a general formula that expresses the phasor $\vec{E}(R)$ in terms of the phasor $\vec{H}(R)$ of a TEM wave and the intrinsic impedance of the medium, where R is the radius vector.

P. 7-1 a) In a source-free conducting medium with constitutive parameters ϵ, μ, and σ,

Eq. (7-62): $\bar{\nabla} \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$

$= \sigma \bar{E} + \epsilon \frac{\partial \bar{E}}{\partial t}$. ①

Eqs. (5-16a) & (7-61): $\bar{\nabla} \times \bar{\nabla} \times \bar{E} = \bar{\nabla} \bar{\nabla} \cdot \bar{E} - \bar{\nabla}^2 \bar{E}$

$= -\mu \frac{\partial}{\partial t} (\bar{\nabla} \times \bar{H})$. ②

Substituting ① in ② and noting that $\bar{\nabla} \cdot \bar{E} = 0$, we obtain the wave equation in dissipative media:

$$\bar{\nabla}^2 \bar{E} - \mu\sigma \frac{\partial \bar{E}}{\partial t} - \mu\epsilon \frac{\partial^2 \bar{E}}{\partial t^2} = 0. \quad ③$$

Similarly for $\bar{H}$.

b) For time-harmonic fields: $\frac{\partial}{\partial t} \rightarrow (j\omega)$ and $\frac{\partial^2}{\partial t^2} \rightarrow (-\omega^2)$.

Wave equation ③ converts to Helmholtz's equation:

$$\bar{\nabla}^2 \bar{E} - j\omega\mu\sigma \bar{E} + k^2 \bar{E} = 0, \quad \text{where } k = \omega\sqrt{\mu\epsilon}.$$

P. 7-3 Assume that $\bar{H}(\bar{R})$ has the form:

$$\bar{H}(\bar{R}) = \bar{H}_0 e^{-jk\bar{a}_k \cdot \bar{R}}.$$

Then, from Eq. (6-50b),

$$\begin{aligned}
\bar{E}(\bar{R}) &= \frac{1}{j\omega\epsilon} \nabla \times \bar{H}(\bar{R}) \\
&= \frac{1}{j\omega\epsilon} (-jk) \bar{a}_k \times \bar{H}(\bar{R}) \\
&= -\frac{1}{\omega\epsilon} (\omega\sqrt{\mu\epsilon}) \bar{a}_k \times \bar{H}(\bar{R}),
\end{aligned}$$

or,

$$\bar{E}(\bar{R}) = -\eta \bar{a}_k \times \bar{H}(\bar{R}).$$

Ex

7.4

The instantaneous expression for the magnetic field intensity of a uniform plane wave propagating in the +y direction in air is given by

$$\vec{H} = \vec{a}_z 4\times10^{-6}\cos\left(10^7\pi t - k_0 y + \frac{\pi}{4}\right) \text{ A/m}$$

a) Determine k_0 and the location where H_z vanishes at $t = 3$ ms.

b) Write the instantaneous expression for $\vec{E}$.

Ex 7.5

The E-field of a uniform plane wave propagating in a dielectric medium is given by

$$E(t,z) = \vec{a}_x 2\cos\left(10^8 t - z/\sqrt{3}\right) - \vec{a}_y \sin\left(10^8 t - z/\sqrt{3}\right) \text{ V/m}$$

a) Determine the frequency and wavelength of the wave

b) What is the dielectric constant of the medium?

c) Describe the polarization of the wave.

d) Find the corresponding H-field

P. 7-4 $\bar{H} = \bar{a}_z\, 4\times10^{-6}\cos(10^7\pi t - k_0 y + \frac{\pi}{4})$ (A/m).

a) $k_0 = \omega\sqrt{\mu_0\epsilon_0} = \frac{10^7\pi}{3\times10^8} = \frac{\pi}{30} = 0.105$ (rad/m).

$\lambda = 2\pi/k_0 = 60$ (m).

At $t = 3\times10^{-3}$ (s), we require the argument of cosine in $\bar{H}$:

$$10^7\pi(3\times10^{-3}) - \frac{\pi}{30}y + \frac{\pi}{4} = \pm n\pi + \frac{\pi}{2}, \quad n = 0, 1, 2, \cdots$$

$$\longrightarrow\quad y = \pm 30n - 7.5\ (\text{m}) = 22.5 \pm n\lambda/2\ (\text{m}).$$

b) Use phasors with cosine reference:

$$\bar{H} = \bar{a}_z\, 4\times10^{-6}\, e^{j(-k_0 y + \pi/4)} \quad (\text{A/m}).$$

From the result of Problem P. 7-3,

$$\bar{E} = -\eta_0\, \bar{a}_y \times \bar{a}_z\, 4\times10^{-6}\, e^{j(-k_0 y + \pi/4)}$$

$$= -\bar{a}_x\, 4\times10^{-6}\,\eta_0\, e^{j(-0.105y + \pi/4)}$$

$$= -\bar{a}_x\, 1.51\times10^{-3}\, e^{j(-0.105y + \pi/4)} \quad (\text{V/m}).$$

The instantaneous expression for $\bar{E}$ is:

$$\bar{E}(y,t) = -\bar{a}_x\, 1.51\cos(10^7\pi t - 0.105y + \pi/4) \quad (\text{mV/m}).$$

P. 7-5 Use phasors with cosine reference.

$$\bar{E}(z) = \bar{a}_x 2 e^{-jz/\sqrt{3}} + \bar{a}_y j e^{-jz/\sqrt{3}} \quad (V/m).$$

a) $\omega = 10^8$ (rad/s) $\longrightarrow$ $f = 10^8/2\pi = 1.59 \times 10^7$ (Hz),

$\beta = 1/\sqrt{3}$ (rad/m) $\longrightarrow$ $\lambda = 2\pi/\beta = 2\sqrt{3}\pi$ (m).

b) $u = \frac{c}{\sqrt{\epsilon_r}} = \frac{\omega}{\beta} \longrightarrow \epsilon_r = \left(\frac{\beta c}{\omega}\right)^2 = 3.$

c) Left-hand elliptically polarized.

d) $\eta = \sqrt{\frac{\mu}{\epsilon}} = \frac{120\pi}{\sqrt{\epsilon_r}} = \frac{120\pi}{\sqrt{3}}$ (Ω),

$$\bar{H} = \frac{1}{\eta}\bar{a}_z \times \bar{E} = \frac{\sqrt{3}}{120\pi}(\bar{a}_y 2 e^{-jz/\sqrt{3}} - \bar{a}_x j e^{-jz/\sqrt{3}}),$$

$$\bar{H}(z,t) = \frac{\sqrt{3}}{120\pi}\left[\bar{a}_x \sin(10^8 t - z/\sqrt{3}) + \bar{a}_y 2\cos(10^8 t - z/\sqrt{3})\right] \quad (A/m).$$

Ex

7.6

Show that a plane wave with an instantaneous expression for the electric field

$$\vec{E}(z,t) = \vec{a}_x E_{10} \sin(\omega t - kz) + \vec{a}_y E_{20} \sin(\omega t - kz + \psi)$$

is elliptically polarized.

P. 7-6 Let $\alpha = \omega t - kz$.

$$\bar{E} = \bar{a}_x E_{10} \sin\alpha + \bar{a}_y E_{20} \sin(\alpha + \psi)$$
$$= \bar{a}_x E_x + \bar{a}_y E_y .$$

$$\frac{E_x}{E_{10}} = \sin\alpha, \quad \frac{E_y}{E_{20}} = \sin(\alpha+\psi)$$
$$= \sin\alpha\cos\psi + \cos\alpha\sin\psi$$
$$= \frac{E_x}{E_{10}}\cos\psi + \sqrt{1-\left(\frac{E_x}{E_{10}}\right)^2}\sin\psi :$$

$$\left(\frac{E_y}{E_{20}} - \frac{E_x}{E_{10}}\cos\psi\right)^2 = \left[1 - \left(\frac{E_x}{E_{10}}\right)^2\right]\sin^2\psi .$$

Rearranging:

$$\left(\frac{E_y}{E_{20}\sin\psi}\right)^2 + \left(\frac{E_x}{E_{10}\sin\psi}\right)^2 - 2\,\frac{E_x E_y}{E_{10}E_{20}}\,\frac{\cos\psi}{\sin^2\psi} = 1,$$

which is the equation of an ellipse in E_x-E_y plane.